

Qatar-1b

R-2421

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_111016 · created 2026-08-21T23:42:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455850.69614 +/- 0.00036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.019
Transit depth (Rp/Rs)^2	2.64 +/- 0.61 [%]
Orbital Inclination (inc)	84.41 +/- 1.21
Airmass coefficient 1 (a1)	1.03 +/- 0.019
Airmass coefficient 2 (a2)	-0.019 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.86 %
Best Comparison Star	#3 - [465, 369]
Optimal Method	PSF photometry
Transit Duration (day)	0.0716 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.162 ± 0.019 vs 0.1463 (deviation 0.83σ)
Duration fit vs published	0.0716 d vs 0.0692 d (deviation 0.4σ)
Mid-time O-C	-0.5 min
Depth SNR	8.5
Residual scatter	1.86 %

Light curve

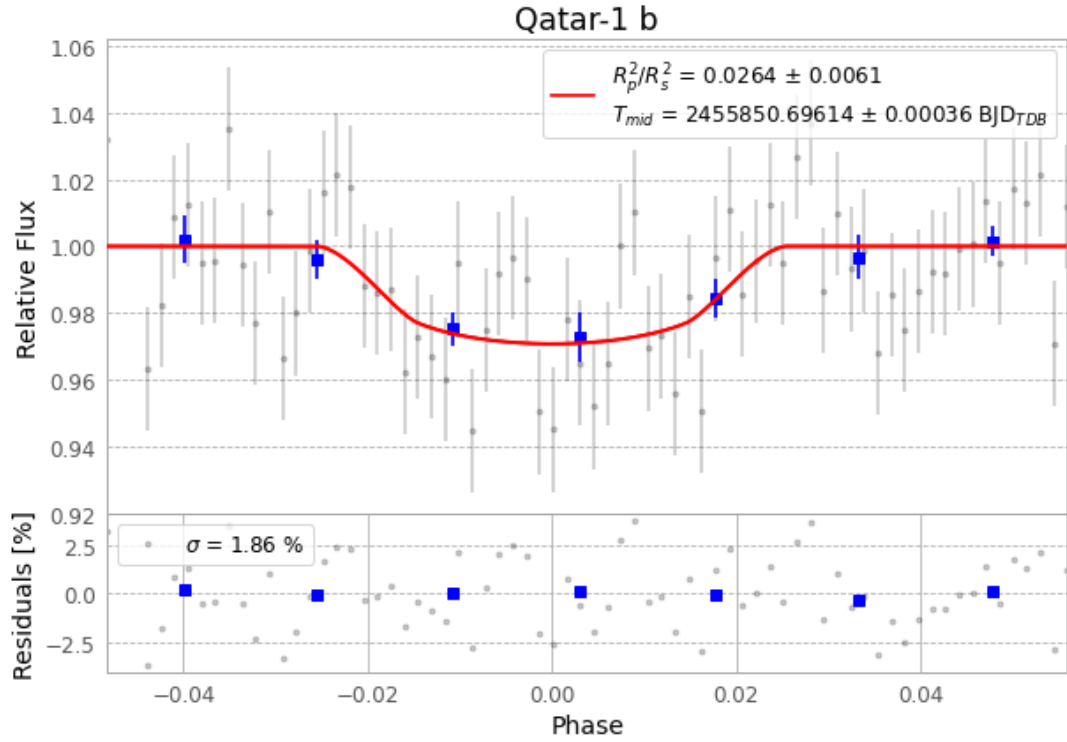


Figure 1 — FinalLightCurve_Qatar-1 b_2011-10-15.png (SHA-256 b21791975c93cae5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 291], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 878b33d6bf8c8165...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1111016030012.FITS → Qatar-1111016063312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-111016.log (02a24bc850d8...), FinalLightCurve_Qatar-1 b_2011-10-15.png (b21791975c93...), FinalLightCurve_Qatar-1 b_2011-10-15.csv (ec43fa420d94...)

Compute resources

Wall time 123.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:42Z.